

Wonjun Lee

PH.D. CANDIDATE IN ELECTRICAL ENGINEERING, CILAB, KAIST

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I am currently pursuing a Ph.D. degree in Electrical Engineering at KAIST. My academic journey began at Gyeonggi Science High School for the Gifted, which fostered my passion for research and innovation. I then earned my B.S. in Electrical Engineering and Computer Science at KAIST, and M.S. in Electrical Engineering at KAIST. I enjoy engaging with creative challenges and value building meaningful relationships along the way. Please feel free to explore my academic background below and contact me directly for further details.

Research Interest

As AI's role expands across various fields, building **trustworthy AI** is crucial. I am particularly interested in AI security, as vulnerabilities compromise reliability and fairness.

Additionally, as humans rely on various senses for judgment, I focus on research in **Multi-Modal Learning** and **AI Security**. My goal is to help bridge the understanding between AI and humans. My work explores the following, but not limited to:

- **Multi-Modal Learning**: multi-modal large language models, efficient multi-modal learning
- **AI Security**: adversarial attack, jailbreaking and their defense
- **Multimedia Forensics**: detecting forgeries, synthetic images, and deepfakes

Education

Korea Advanced Institute of Science and Technology (KAIST)

PH.D. IN ELECTRICAL ENGINEERING

Daejeon, S.Korea

Mar. 2025 - Present

Korea Advanced Institute of Science and Technology (KAIST)

M.S. IN ELECTRICAL ENGINEERING

Daejeon, S.Korea

Mar. 2023 - Feb. 2025

Korea Advanced Institute of Science and Technology (KAIST)

B.S. IN ELECTRICAL ENGINEERING & SCHOOL OF COMPUTING (DOUBLE MAJOR)

Daejeon, S.Korea

Mar. 2019 - Feb. 2023

Gyeonggi Science High School for the Gifted (GSHS)

MAJOR IN MATH, PHYSICS, AND BIOLOGY

Suwon, Gyeonggi-do, S.Korea

Mar. 2016 - Feb. 2019

Publication

INTERNATIONAL

Safety in Batches? Understanding and Mitigating Safety Failures in Batch Prompting

ASIA-PACIFIC CHAPTER OF ACL & INTERNATIONAL JOINT CONFERENCE ON NLP (AACL-IJCNLP MAIN)

Kihyun Kim, Hee-Seon Kim, **Wonjun Lee**, Changick Kim

2026

LLM Safety

AIM: Anchor Identity Features, then Match for Multimodal Large Language Model Unlearning

THE 2026 CONFERENCE ON EMPIRICAL METHODS IN NATURAL LANGUAGE PROCESSING (EMNLP FINDINGS)

Wonjun Lee*, Jaehyuk Jang*, Kangwook Ko*, Hee-Seon Kim, Changick Kim (*: Equal Contribution)

Where Identity Lives: Localized, Retain-Free Identity Unlearning in Multimodal Large Language Models

THE 2026 CONFERENCE ON EMPIRICAL METHODS IN NATURAL LANGUAGE PROCESSING (EMNLP FINDINGS)

Kangwook Ko*, Jaehyuk Jang*, **Wonjun Lee***, Hee-Seon Kim, Changick Kim (*: Equal Contribution)

SubT: Subspace Tuning for Few-shot Generalization of Audio-Language Models

THE 2026 CONFERENCE ON EMPIRICAL METHODS IN NATURAL LANGUAGE PROCESSING (EMNLP FINDINGS)

Jaehyuk Jang, Kangwook Ko, **Wonjun Lee**, Changick Kim

2026

Multimodal LLM Unlearning

2026

Multimodal LLM Unlearning

2026

ALM Prompt Tuning

Jailbreak to Protect: Buffering Harmful Fine-Tuning via Temporary Jailbreaking LoRA in Large Language Models

INTERNATIONAL CONFERENCE ON MACHINE LEARNING (ICML) **SPOTLIGHT (TOP 2.2%)**

Seokil Ham, Jaehyuk Jang, **Wonjun Lee**, Changick Kim

2026

Multimodal LLM Safety

Generalizable Prompt Tuning for Audio-Language Models via Semantic Expansion

THE 64TH ANNUAL MEETING OF THE ASSOCIATION FOR COMPUTATIONAL LINGUISTICS (ACL FINDINGS)

Jaehyuk Jang*, **Wonjun Lee***, Kangwook Ko*, Changick Kim (*: Equal Contribution)

2026

ALM Prompt Tuning

Efficient Test-Time Optimization for Depth Completion via Low-Rank Decoder Adaptation

PREPRINT

Minseok Seo*, **Wonjun Lee***, Jaehyuk Jang, Changick Kim (*: Equal Contribution)

2026
Depth Estimation, VFM

SELF: Selective Fusion of Identity for Generalizable Deepfake Detection

PREPRINT

Younghun Kim, Minsuk Jang, Myung-Joon Kwon, **Wonjun Lee**, Changick Kim

2025
Deepfake Detection

Benign-to-Toxic Jailbreaking: Inducing Harmful Responses from Harmless Prompts

PREPRINT

Hee-Seon Kim, Minbeom Kim, **Wonjun Lee**, Kihyun Kim, Changick Kim

2025
Multimodal LLM Safety

SAFIRE: Segment Any Forged Image Region

AAAI CONFERENCE ON ARTIFICIAL INTELLIGENCE (AAAI)

Myung-Joon Kwon*, **Wonjun Lee***, Seung-Hun Nam, Minji Son, Changick Kim (*: Equal Contribution)

2025
Image Forgery Localization

FRIDAY: Mitigating Unintentional Facial Identity in Deepfake Detectors Guided by Facial Recognizers

VISUAL COMMUNICATIONS AND IMAGE PROCESSING (VCIP) **ORAL PRESENTATION**

Younghun Kim, Myung-Joon Kwon, **Wonjun Lee**, Changick Kim

2024
Deepfake Detection

DOMESTIC

Image-Text Order Matters: Input Ordering Effects in LVLM Jailbreaks

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE) **LG ELECTRONICS PAPER AWARD**

Kihyun Seol, **Wonjun Lee**, Kihyun Kim, Minbeom Kim, Changick Kim

2026
Multimodal LLM Safety

Safety Analysis of Image Verification-Based Reasoning Methods in MLLMs

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE)

Jihyun Lee, **Wonjun Lee**, Kihyun Kim, Hee-Seon Kim, Changick Kim

2026
Multimodal LLM Safety

Image-Based Universal Jailbreak Attack via Safety Boundary Approximation

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE)

Kihyun Kim, Hee-Seon Kim, Minbeom Kim, **Wonjun Lee**, Changick Kim

2025
Multimodal LLM Safety

Ensembling Pretrained Models for Enhanced Generalization in Synthetic Image Detection

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE) **LG ELECTRONICS PAPER AWARD**

Myung-Joon Kwon, **Wonjun Lee**, Younghun Kim, Changick Kim

Synthetic Image Detection

Combating Bias: Corrective Measures Through Adversarial Training in Neural Networks

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE) **LG ELECTRONICS PAPER AWARD**

Wonjun Lee, Myung-Joon Kwon, Younghun Kim, Changick Kim

2024
De-Biasing, Adversarial Training

Improving the Generalization Performance of Deepfake Detection Through a Partial Image Manipulation Augmentation Technique

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE)

Younghun Kim, Myung-Joon Kwon, **Wonjun Lee**, Changick Kim

2024
Deepfake Detection

Research Experience

Roen Surgical

KAIST EE EXTERNSHIP PROGRAM SECOND COHORT

- Participated as a key developer in the patent development of "Stone size estimation method."
- Contributed to technical development and validation for image-based stone size estimation in surgical applications.

Daejeon, S.Korea
Jun.2022 – Dec.2022

SAMSUNG Electronics CE/IM Division Mobile Communications Unit

SUMMER INTERNSHIP

- Worked on Galaxy Tab FE design.
- Gained hands-on experience with mobile wireless communication module design and product development workflows.

Remote
Jul.2021 - Aug.2021

SAMSUNG Electronics DS Division Foundry Business Unit

SAMSUNG TALENT INTERNSHIP PROGRAM (STIP)

- Worked on DRAM design.
- Gained hands-on experience with semiconductor memory design flows and foundry development workflows.

Hwaseong, Gyeonggi-do, S.Korea
Jul.2020 - Aug.2020

Patent

A Universal Image-Generation Framework for Jailbreaking Large Vision-Language Models by Bypassing Safety Alignment

2025

KR PATENT 10-2025-0157353

- Participated as a Key Developer in Patent Development
- Participated through the Project with ETRI

Method and System for Generating Universal Adversarial Perturbations Using High-Sensitivity Components of Vision Encoders in Large-Scale Vision-Language Models

2025

KR PATENT 10-2025-0194468

- Participated as a Key Developer in Patent Development
- Participated through the Project with IITP

Stone size estimation method

2023

KR PATENT 10-2023-0031846

- Participated as a Key Developer in Patent Development
- Participated through the KAIST EE Externship

Honors & Awards

2026 **LG Electronics Paper Award**, Summer Annual Conference of IEIE

Jeju, S.Korea

2024 **LG Electronics Paper Award (2 papers)**, Summer Annual Conference of IEIE

Jeju, S.Korea

2023 **Certificate**, KAIST Leadership and Volunteer Excellence Award Certificate

Daejeon, S.Korea

2018 **Bronze Award**, SAMSUNG Humantech Paper Award Biology Division

Seoul, S.Korea

Scholarship

Korea National Scholarship (B.S.)

Mar. 2019 - Feb. 2023

Korea National Scholarship (M.S.)

Mar. 2023 - Feb. 2025

Korea National Scholarship (Ph.D.)

Mar. 2025 - Present

Project

Automatic Evaluation Benchmark Generation from LLM Log Data

Jun. 2025 - Aug. 2030

SAMSUNG ELECTRONICS DS

Development of AI Technology with Robust and Flexible Resilience Against Risk Factors

Jan. 2025 - Dec. 2028

ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE (ETRI)

- Team Leader

Penetration Security Testing of ML Model Vulnerabilities and Defense

Jan. 2025 - Dec. 2027

INSTITUTE FOR INFORMATION & COMMUNICATION TECHNOLOGY PLANNING & EVALUATION (IITP)

Scene Text Recognition with Visual Contexts

Sep. 2024 - Dec. 2024

CENTER FOR SECURITY TECHNOLOGY RESEARCH, KAIST

Bypass Techniques for Identifying Vulnerabilities in CAPTCHA

Mar. 2023 - Dec. 2023

CENTER FOR SECURITY TECHNOLOGY RESEARCH, KAIST

Practical Adversarial Attacks of AI Facial Recognition Technology Using Physical Patterns

Mar. 2023 - Dec. 2023

CENTER FOR SECURITY TECHNOLOGY RESEARCH, KAIST

Academic Service

Visual Communications and Image Processing (VCIP)

2025 - Present

AAAI Conference on Artificial Intelligence (AAAI)

2026 - Present

IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)

2026 - Present

KAIST Electrical Engineering TA MEMBER • Signals and Systems - Fall 2023 • Introduction to Electronics Design Lab - Fall 2023, Fall 2024, Spring 2025, Spring 2026 • Digital Image Processing - Spring 2024 • EE Career Development I - Spring 2025, Spring 2026 • Introduction to Machine Learning - Fall 2025	<i>Daejeon, S.Korea</i> <i>Sep. 2023 - Present</i>
KAIST Student Counseling Assistant (CA) MEMBER • The CA (Counseling Assistant) program aims to expand counseling support opportunities for students and graduate students to ensure a smooth academic/school life and career planning. • Participants work autonomously with peers to enhance their development skills and foster personal growth.	<i>Daejeon, S.Korea</i> <i>Mar. 2024 - Feb. 2025, Sep. 2025 - Present</i>
CILAB, KAIST SERVER VICE TEAM LEADER VICE LAB LEADER SERVER TEAM LEADER	<i>Daejeon, S.Korea</i> <i>Nov. 2025 - Nov. 2026</i> <i>Nov. 2025 - Nov. 2026</i> <i>Nov. 2024 - Nov. 2025</i>
KAIST Electrical Engineering Student Council VICE DEPARTMENT REPRESENTATIVE	<i>Daejeon, S.Korea</i> <i>Feb. 2020 - Feb.2022</i>
KAIST Dance Crew Illusion PRESIDENT	<i>Daejeon, S.Korea</i> <i>Jan. 2020 - Feb.2022</i>
MadCamp 11th Cohort Hosted by KAIST CS MEMBER • MadCamp is an intensive program in collaboration with KAIST, focused on app, web, and game development for undergraduate students in South Korea. • Participants work autonomously with peers to enhance their development skills and foster personal growth.	<i>Daejeon, S.Korea</i> <i>Dec. 2020 - Jan. 2021</i>